Game Development Technologies Research Workbook

Game Engines

1. Identify 3 different game engines that can be used to create 3D video games.
   Research each game engine and provide a list of features (at least 5) for each engine.
   One of these engines must be the Unreal Engine 4.

Game Engine	Features
Unreal Engine 4	Robust multiplayer support
	C++
	In engine rendering
	Animation blueprints
	Scalable foliage
Unity	C#
	Frame debugger
	Visual effects
	Input
	UI
Godot	Building blocks
	Data driven elements
	C#
	Community supported languages
	Both high and low end devices

2. What makes each engine different?
   Some of the features you identified may be common across two or more engines. For each game engine identified, write a short statement about what makes this engine unique and why a game studio may want to use it. (Write one or two sentences for each engine)

Game Engine	Unique Selling Point
Unreal Engine 4	Unreal engine hosts an easy to learn and use system, which still provides professional tools.
Unity	Unity has high capability when creating crossplatform games, as well as a massively supportive community.
Godot	Godot's unique selling point is being completely free with a lot of community support.

3. For each game engine identified, list sources of information for developers to refer to. This can be online technical information provided by the developer, how-to articles and blogs, youtube channels, or textbooks.
   Identify at least 2 different sources for each engine.

Game Engine	Sources of Information
Unreal Engine 4	Epic Developer Community Forums
	https://www.youtube.com/channel/UCBobmJyzsJ6Ll7UbfhI4iwQ
Unity	Unity Discussions - A Space to Discuss All Things Unity

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	https://www.youtube.com/c/unity
Godot	https://forum.godotengine.org/
	https://www.youtube.com/c/GodotTutorials

Image Editing Software

4. Identify 3 different programs used for image editing.
   These may be PC applications, web services, or mobile apps.
   Research each program and provide a list of features (at least 3) for each.
   One of these programs must be Krita.

Image Editing Software	Features
Krita	Clean and flexible interface
	Wrap around mode
	Vector and text
Photopea	Background removal
	Layers
	Blending modes
Adobe photoshop	Stylised text
	Combine pictures
	Change any colour of an object

5. Identify a unique selling point for each image editing program. This might include its licensing or cost, its feature set, ease-of-use, or something else. (Write one or two sentences for each program)

Image Editing Software	Unique Selling Point
Krita	Krita provides professional tools in a unique and customisable way.
Photopea	Photopea is free to use and does not require a download in anyway
Adobe photoshop	Adobe photosop hosts industry standard image editing software as well as generative AI.

6. For each image editing program identified, list sources of information for users to refer to. This can be online technical information provided by the developer, how-to articles and blogs, youtube channels, or textbooks.
   Identify at least 2 different sources for each engine.

Image Editing Software	Sources of Information
Krita	Krita | Digital Painting. Creative Freedom.
	Krita Artists - Krita community forum
photopea	Photopea - Introduction
	https://www.youtube.com/playlist?list=PLO3K3VFvlU6D1KgYzMtAyFEBR3mkxl9Yw
Adobe photoshop	Explore New Features in Photoshop | Adobe | Adobe Australia
	https://www.youtube.com/@Photoshop

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Audio Software

7. Identify 3 different programs used for editing audio.
   These may be PC applications, web services, or mobile apps.
   Research each program and provide a list of features (at least 3) for each.
   One of these programs must be Audacity.

Audio Editing Software	Features
Audacity	Pitch shifter
	Noise reduction
	Vocal remover
Adobe Audition	Royalty free sound effects
	Impressive texture adders
	Eliminate white noise
FL studio	FLEX
	Effects
	Deep tools

8. Identify a unique selling point for each audio editing program. This might include its licensing or cost, its feature set, ease-of-use, or something else. (Write one or two sentences for each program)

Audio Editing Software	Unique Selling Point
Audacity	Audacity's unique selling point is that it provides a useful and completely free software.
Adobe audition	Adobe audition's unique selling point is that it is an all purpose audio editor.
FL studio	FL studio's unqieu sellingpoint is that it has lifetime free updates, and a non-linear workflow.

9. For each audio editing program identified, list sources of information for users to refer to. This can be online technical information provided by the developer, how-to articles and blogs, youtube channels, or textbooks.
   Identify at least 2 different sources for each engine.

Audio Editing Software	Sources of Information
Audacity	Features – Audacity
	https://forum.audacityteam.org/
Adobe audition	New features summary for the latest release of Adobe Audition.
	NEW Audition Support Community - Welcome! | Community
FL studio	Features | FL Studio
	https://forum.image-line.com/

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